8.03SC Problem Set 7

Nilangshu Sarkar

February 2026

Problem 7.1

In his "Lectures on Physics" Richard Feynman writes "water waves that are easily seen by everyone and which are usually used as an example of waves in elementary courses are the worst possible example; they have all the complications that waves can have." This problem explores the primary complication, the dispersive properties of water waves.

Consder waves on the surface of water (also known as surface gravity waves). Assume that the deformations of the surface are influenced by gravity g (on Earth) and the surface tension T of the water (expressed in units of Newton per meter). If the density of the water is ρ and the undisturbed depth of the water is h, one can show that the phase velocity of the surface waves is given by

$$v_p^2(k) = \left(\frac{g}{k} + \frac{Tk}{\rho}\right)\tanh(kh)$$

where $k = 2\pi/\lambda$ is the wavennumber, $\tanh(kh)$ is the hyperbolic tangent, and viscosity is neglected. For water $\rho = 1$ gcm^{-3}

a. In terms of the given quantities, at approximately what wavelength $\lambda_{\text{critical}}$ does the effect of surface tension become comparable to the effect of gravity? For water at a temperature of 20°C the critical wavelength is $\lambda_{\text{critical}} = 2$ cm. What is the surface tension T of water?

b. Suppose $\lambda \gg \lambda_{\text{critical}}$ so that surface tension is negligible. What is the phase velocity and group velocity for shallow water waves (wavelength $\lambda \gg h$)? Are the waves dispersive under this condition?

c. Suppose $\lambda \gg \lambda_{\text{critical}}$ so that surface tension is negligible. What is the phase velocity and group velocity for deep water waves ($\lambda \ll h$)? Are the waves dispersive under this condition?

d. Suppoe that the water is deep $\lambda \ll h$ and the wavelengths are so short that $\lambda \ll \lambda_{\text{critical}}$ such that surface tension dominates and gravity is negligible. Such waves are called capillary waves. What is their phase velocity and group velocity? Are capillary waves dispersive?

a.

Surface tension and gravity contribute equally when

$$\frac{g}{k} \sim \frac{Tk}{\rho}$$

Hence,

$$g = \frac{T}{\rho}k^2 \quad \Rightarrow \quad k_c = \sqrt{\frac{\rho g}{T}}$$

The corresponding wavelength is

$$\boxed{\lambda_c = \frac{2\pi}{k_c} = 2\pi\sqrt{\frac{T}{\rho g}}}$$

Solving for T,

$$T = \frac{\rho g \lambda_c^2}{4\pi^2}$$

Using

$$\rho = 10^3\,\mathrm{kg\ m^{-3}}, \qquad g = 9.8\,\mathrm{m\ s^{-2}}, \qquad \lambda_c = 2\,\mathrm{cm} = 2\times 10^{-2}\,\mathrm{m}$$

we obtain

$$\boxed{T \approx 0.072\,\mathbf{N/m}}$$

which agrees with the known surface tension of water.

b.

Surface tension is negligible:

$$v_p^2 = \frac{g}{k}\tanh(kh)$$

For shallow water ($kh \ll 1$),

$$\tanh(kh) \approx kh$$

Thus,

$$v_p^2 = gh$$

Phase Velocity

$$\boxed{v_p = \sqrt{gh}}$$

The dispersion relation becomes

$$\omega = v_p k = \sqrt{gh}\,k$$

Group velocity:

$$\boxed{v_g = \frac{d\omega}{dk} = \sqrt{gh}}$$

Therefore,

$$v_g = v_p$$

and the waves are non-dispersive.

c.

Here $\tanh(kh) \approx 1$, giving

$$v_p^2 = \frac{g}{k}$$

Hence

$$\boxed{v_p = \sqrt{\frac{g}{k}}}$$

Since $\omega = v_p k$,

$$\omega = \sqrt{gk}$$

Group velocity:

$$\boxed{v_g = \frac{d\omega}{dk} = \frac{1}{2}\sqrt{\frac{g}{k}} = \frac{1}{2}v_p}$$

Thus the waves are dispersive.

d.

Gravity is negligible and $\tanh(kh) \approx 1$:

$$v_p^2 = \frac{Tk}{\rho}$$

Therefore

$$\boxed{v_p = \sqrt{\frac{T}{\rho}}\, k^{1/2}}$$

The dispersion relation becomes

$$\omega = v_p k = \sqrt{\frac{T}{\rho}}\, k^{3/2}$$

Group velocity:

$$\boxed{v_g = \frac{d\omega}{dk} = \frac{3}{2}\sqrt{\frac{T}{\rho}}\, k^{1/2} = \frac{3}{2}v_p}$$

Hence capillary waves are also dispersive.

Problem 7.3

Consider an infinite string of mass density ρ_L under tension T. A Gaussian shaped pulse is traveling on the string, causing a vertical displacement given by:

$$y(x,t) = exp\left[-\frac{1}{2}\left(\frac{(x-vt)}{\sigma}\right)^2\right]$$

a. Find the distribution of frequencies $C(\omega)$ that contribute to this pulse by calculating the "'inverse Fourier transform" (eq 10.37 in Georgi). This is a very well known integral, $C(\omega)$ is also a Gaussian with width σ_ω, feel free to use online tools to find it. Make sure that all the constants are correct.

b. Calculate the product of the widths of the two Gaussians, $(\sigma \cdot \sigma_\omega)$.

a.

At $x = 0$,

$$y(0,t) = \exp\left[-\frac{1}{2}\left(\frac{vt}{\sigma}\right)^2\right] = \exp\left[-\frac{t^2}{2\tau^2}\right]$$

where

$$\tau = \frac{\sigma}{v}$$

The inverse Fourier transform is

$$C(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} y(0,t)\, e^{i\omega t}\, dt$$

Substituting the Gaussian,

$$C(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \exp\left(-\frac{t^2}{2\tau^2} + i\omega t\right) dt$$

Using the standard Gaussian integral

$$\int_{-\infty}^{\infty} e^{-at^2+bt}\, dt = \sqrt{\frac{\pi}{a}}\, e^{b^2/(4a)}$$

with

$$a = \frac{1}{2\tau^2}, \qquad b = i\omega$$

we obtain

$$C(\omega) = \tau \exp\left(-\frac{\omega^2\tau^2}{2}\right)$$

Restoring $\tau = \sigma/v$,

$$\boxed{C(\omega) = \frac{\sigma}{v}\exp\left[-\frac{1}{2}\left(\frac{\sigma\omega}{v}\right)^2\right]}$$

Thus the frequency distribution is also a Gaussian.
The frequency width is therefore

$$\sigma_\omega = \frac{v}{\sigma}$$

b.

The spatial width of the pulse is σ and the frequency width is

$$\sigma_\omega = \frac{v}{\sigma}$$

Hence

$$\boxed{\sigma\,\sigma_\omega = v}$$

Physical Interpretation

A narrow pulse in space requires a broad spread of frequencies, while a wide pulse corresponds to a narrow frequency spectrum:

$$\boxed{\sigma \uparrow \Rightarrow \ \sigma_\omega \downarrow \qquad \sigma \downarrow \Rightarrow \ \sigma_\omega \uparrow}$$

This inverse relation is the classical wave analogue of an uncertainty relation and reflects the Fourier duality between localization in space and localization in frequency.